



CMPE 472 – Computer Networks

Lab 1: HTTP (3 points)

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This lab assignment will help you gain hands-on experience in using Wireshark to analyze HTTP traffic, which is crucial for network administrators, security analysts, and web developers. It will also improve your understanding of how the HTTP protocol works in real-world scenarios. The goal of this lab assignment is to familiarize you with Wireshark and how it can be used to capture and analyze HTTP traffic.

Download the `bbc_news.pcapng` file, which captures the network traffic after browsing the news on BBC, to your computer via LMS.

And open this file through Wireshark and answer the following questions through this file. Please fill out this given form for your submissions.

Add the necessary screenshots showing where you find that answer.

Questions

1. Fill in the table below by examining the line of the server's **first HTTP response** to the client. (0.5 points)

Client IP Address	<i>195.175.178.203</i>
Website IP Address	<i>192.168.1.112</i>
Protocol Version	<i>TCP (6)</i>
Return Status	<i>200 OK</i>
Content Type	<i>application/octet-stream\r\n</i>

http						
No.	Time	Source	Destination	Protocol	Length	Info
5637	8.830807	192.168.1.112	195.175.178.203	HTTP	370	SET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/pieceshash HTTP/1.1
5658	8.841325	195.175.178.203	192.168.1.112	HTTP	531	HTTP/1.1 200 OK
5677	8.910347	192.168.1.112	195.175.178.138	HTTP	475	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
5680	8.913962	192.168.1.112	195.175.178.121	HTTP	475	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
5682	8.958195	195.175.178.138	192.168.1.112	HTTP	910	HTTP/1.1 206 Partial Content
5684	8.958195	195.175.178.121	192.168.1.112	HTTP	910	HTTP/1.1 206 Partial Content
5685	8.959156	192.168.1.112	195.175.178.121	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
5686	8.959900	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
6156	9.249667	195.175.178.138	192.168.1.112	HTTP	251	HTTP/1.1 206 Partial Content
6165	9.251790	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
6281	9.324612	195.175.178.121	192.168.1.112	HTTP	663	HTTP/1.1 206 Partial Content
6292	9.326529	192.168.1.112	195.175.178.121	HTTP	481	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
7094	9.537336	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF
7097	9.582747	192.168.1.112	195.175.178.121	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c?P1=17105913748P2=404&P3=2&P4=FY47gxzjwZxLmfYV4WuQouQvKruF

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> Ethernet II, Src: Intel_33:f5:86 (ec:63:d7:33:f5:86), Dst: zte_41:87:2d (e8:43:68:41:87:2d) > Internet Protocol Version 4, Src: 192.168.1.112, Dst: 195.175.178.203	
0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) 0000 00.. = Differentiated Services Codepoint: Default (0)00 = Explicit Congestion Notification: Not ECN-Capable Transport (0) Total Length: 356 Identification: 0x6d74 (28020) > 010. = Flags: 0x2, Don't fragment 0... = Reserved bit: Not set .1.. = Don't fragment: Set ..0. = More fragments: Not set ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 128 Protocol: TCP (6)	<pre> 0000 e8 43 68 41 87 2d e 0010 01 64 6d 74 40 00 8 0020 b2 cb cf 6e 00 50 4 0030 02 04 39 ea 00 00 4 0040 74 72 65 61 6d 69 6 0050 66 69 6c 65 73 2f 3 0060 65 36 31 2d 34 64 3 0070 32 31 30 30 65 3 3 0080 73 68 61 73 68 20 4 0090 43 6f 6e 6e 65 63 7 0100 2d 41 6c 69 76 65 0 0110 2a 2f 2a 0d 0a 41 6 0120 64 69 6e 67 3a 20 6 0130 61 74 65 0d 0a 55 7 0140 20 4d 69 63 72 6f 7 0150 65 72 79 2d 4f 70 7 0160 2f 31 30 2e 30 0d 0 0170 46 2f 72 4f 62 4a 3 0180 4c 62 71 41 2e 30 2 0190 45 6f 6e 74 65 6e 7 0200 30 0d 0a 48 6f 73 7 0210 66 2e 64 6c 2e 64 6 0220 2e 6d 69 63 72 6f 7 0230 0d 0a </pre>

- > Hypertext Transfer Protocol
 - > HTTP/1.1 200 OK\r\n
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Cache-Control: public, max-age=17280000\r\n
 - Content-Type: application/octet-stream\r\n

2. Inspect the contents of the **second HTTP GET** request from client's browser to the server. Is there any response, if any what is the frame number of it? (0.5 points)

5682

bbc_news.pcapng

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http

No.	Time	Source	Destination	Protocol	Length	Info
5637	8.830807	192.168.1.112	195.175.178.203	HTTP	370	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/pieceshash HTTP/1.1
5650	8.843325	195.175.178.203	192.168.1.112	HTTP	531	HTTP/1.1 200 OK
5677	8.910347	192.168.1.112	195.175.178.138	HTTP	475	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
5680	8.913962	192.168.1.112	195.175.178.121	HTTP	475	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
5682	8.958195	195.175.178.138	192.168.1.112	HTTP	910	HTTP/1.1 206 Partial Content
5684	8.958195	195.175.178.121	192.168.1.112	HTTP	910	HTTP/1.1 206 Partial Content
5685	8.959156	192.168.1.112	195.175.178.121	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
5686	8.959900	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
6156	9.249667	195.175.178.138	192.168.1.112	HTTP	251	HTTP/1.1 206 Partial Content
6165	9.251790	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
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7094	9.537336	192.168.1.112	195.175.178.138	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
7097	9.582257	192.168.1.112	195.175.178.121	HTTP	489	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
7893	8.809969	195.175.178.138	192.168.1.112	HTTP	1160	HTTP/1.1 206 Partial Content
7903	8.812238	192.168.1.112	195.175.178.138	HTTP	487	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr
7906	8.856289	195.175.178.121	192.168.1.112	HTTP	397	HTTP/1.1 206 Partial Content
7908	8.859154	192.168.1.112	195.175.178.121	HTTP	487	GET /filestreamingservice/files/7bf64e56-5e61-4d5b-aaaa-9d2100e52c3c/P1-17105913748P2-4048P3-28P4-fy47gxzjwZxLmfYV4hQouQvKnr

0101 ... = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

0000 ... = Reserved: Not set

...0 ... = Accurate ECN: Not set

...0 ... = Congestion Window Reduced: Not set

...0 ... = ECN-Echo: Not set

...0 ... = Urgent: Not set

...1 ... = Acknowledgment: Set

...1 ... = Push: Set

...0 ... = Reset: Not set

...0 ... = Syn: Not set

...0 ... = Fin: Not set

[TCP Flags:AP...]

Window: 501

[Calculated window size: 64128]

[Window size scaling factor: 128]

Checksum: 0x2247 [unverified]

3. Measure the response time between the **third HTTP GET** request and its response, and specify the duration in milliseconds. (0.5 points)

0.044233

bbc_news.pcapng

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4. Now inspect the contents of the **forth HTTP GET** request and compare it to the **first get**. Explain the differences. (0.5 points)

Frame number, Frame length, Capture length, Identification, Destination address

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Wireshark - Paket 5685 - bbc_news.pcapng

Field	Value
Section number:	1
Interface id:	0 (\Device\NPF_{9C86E83B-A8B6-44FA-94C7-6B5E48989BC8})
Interface name:	\Device\NPF_{9C86E83B-A8B6-44FA-94C7-6B5E48989BC8}
Interface description:	Wi-Fi
Encapsulation type:	Ethernet (1)
Arrival Time:	Mar 9, 2024 15:17:35.175545000 Türkiye Standart Saati
UTC Arrival Time:	Mar 9, 2024 12:17:35.175545000 UTC
Epoch Arrival Time:	1709986655.175545000
[Time shift from this packet: 0.000000000 seconds]	
[Time delta from previous captured frame: 0.000107000 seconds]	
[Time delta from previous displayed frame: 0.000000000 seconds]	
[Time since reference or first frame: 8.830807000 seconds]	
Frame Number:	5637
Frame Length:	370 bytes (2960 bits)
Capture Length:	370 bytes (2960 bits)
[Frame is marked: False]	
[Frame is ignored: False]	
[Protocols in frame: eth:ethertype:ip:tcp:http]	
[Coloring Rule Name: HTTP]	
[Coloring Rule String: http tcp.port == 80 http2]	
Differentiated Services Field:	0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)	
....00.. = Explicit Congestion Notification: Not ECN-Capable Transport (0)	
Total length:	356
Identification:	0x5d74 (23820)
Section number:	1
Interface id:	0 (\Device\NPF_{9C86E83B-A8B6-44FA-94C7-6B5E48989BC8})
Interface name:	\Device\NPF_{9C86E83B-A8B6-44FA-94C7-6B5E48989BC8}
Interface description:	Wi-Fi
Encapsulation type:	Ethernet (1)
Arrival Time:	Mar 9, 2024 15:17:35.303894000 Türkiye Standart Saati
UTC Arrival Time:	Mar 9, 2024 12:17:35.303894000 UTC
Epoch Arrival Time:	1709986655.303894000
[Time shift from this packet: 0.000000000 seconds]	
[Time delta from previous captured frame: 0.000961000 seconds]	
[Time delta from previous displayed frame: 0.000961000 seconds]	
[Time since reference or first frame: 8.955156000 seconds]	
Frame Number:	5685
Frame Length:	489 bytes (3912 bits)
Capture Length:	489 bytes (3912 bits)
[Frame is marked: False]	
[Frame is ignored: False]	
[Protocols in frame: eth:ethertype:ip:tcp:http]	
[Coloring Rule Name: HTTP]	
[Coloring Rule String: http tcp.port == 80 http2]	
Total Length:	475
Identification:	0x3226 (12838)
0101 ... = Flags: 0x2, Don't fragment	
...0 ... = Reserved bit: Not set	
...1 ... = Don't fragment: Set	

Destination Address: 195.175.178.121
[Stream index: 59]
Transmission Control Protocol, Src Port: 53105, Dst Port: 80, Seq: 422, Ack: 857, Len: 435
Source Port: 53105
Destination Port: 80

[illegible]